Global Electric Vehicle Market Analysis

AnInteractiveTableauDashboardon Sales, Manufacturers,Batteries,Range,Ratings and CO Savings

2

Name	Vishesh Yadav
Enrollment No.	EN25CS3011065
Course	B.Tech CS
Project	Data Analytics with Tableau - Capstone Project

1. Introduction

1.1 Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the automotive market, driven by environmental awareness, government incentives, better battery technology and demand for sustainable transport. The industry produces large volumes of data on sales, pricing, battery specifications, manufacturing locations, customer ratings and sustainability.

This project turns that data into an interactive Tableau Public dashboard of nine visualisations. The dataset (ev_market_test.csv) has 100,000 vehicle records and 37 columns covering 200 manufacturers, 15 vehicle categories, 5 battery technologies, 10 manufacturing countries and 7 market regions, with model years from 2018 to 2026. The dashboard serves three stakeholder scenarios: a market strategy director, a business intelligence manager and a sustainable mobility researcher.

1.2 Purpose

- Show which vehicle categories and manufacturers contribute most to annual sales.
- Compare battery technologies, battery capacity and driving range.
- Show customer satisfaction, price and rating patterns across vehicle segments.
- Show manufacturing distribution by country and estimated CO_2 savings by category.
- Give each of the three scenarios clear, data-backed answers in one dashboard.

2. Ideation Phase

2.1 Problem Statement

No.	Stakeholder	Problem	What they need
1	Emily Johnson, EV Market Strategy Director	Must decide where to focus the product portfolio but cannot easily see which categories, manufacturers and segments perform best.	Category sales, leading manufacturers, customer satisfaction, growth over model years.
2	David Chen, Automotive Business Intelligence Manager	Must advise clients entering the EV market but lacks a single view of manufacturer performance, battery mix and manufacturing geography.	Manufacturer sales, battery distribution, manufacturing country analysis, ratings.
3	Dr. Sarah Williams, Sustainable Mobility Research Analyst	Needs evidence on how batteries, range and manufacturing influence EV adoption and carbon reduction.	Battery comparison, range analysis, manufacturing distribution, CO_2 savings.

2.2 Empathy Map Canvas

Stakeholder	Says	Thinks	Does	Feels
Emily Johnson	Which segments should we back?	Are we missing a high-growth category?	Reviews sales and satisfaction by segment, plans the portfolio.	Pressure to justify investment decisions.
David Chen	Where can my client compete?	Who leads, and which battery type dominates?	Benchmarks manufacturers, studies production geography.	Needs clear proof for client recommendations.
Dr. Sarah Williams	Which technologies drive adoption?	Does more capacity really mean more range?	Compares batteries and vehicle types, studies CO_2 impact.	Curious but careful about causal claims.

2.3 Brainstorming

Candidate visuals were listed for each scenario and shortlisted by how directly they answer the scenario questions. The nine selected views are below.

Idea	Scenario	Decision
Sales by vehicle category	1	Selected
EV sales by model year	1	Selected
Customer satisfaction by category	1	Selected
Annual sales by manufacturer	2 (and 1)	Selected
Manufacturing distribution by country	2 (and 3)	Selected
Price vs customer rating	2	Selected
Battery technology distribution	3 (and 2)	Selected
Battery capacity vs driving range	3	Selected
CO savings by category	3	Selected

2

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Dashboard touchpoint	Outcome
Discover	Opens the project website or dashboard link	Website overview and embedded dashboard	Understands the scope at a glance.
Explore	Scans the nine views	Views grouped by scenario	Finds the views that answer their own question.
Analyse	Hovers over charts, compares segments	Tooltips, colours, sorted axes	Compares categories, manufacturers, batteries and countries.
Decide	Draws a conclusion	Key findings in Section 7	Chooses a segment, a client strategy or a research direction.
Share	Shares the link or report	Tableau Public link, GitHub repository, website	Others see the same evidence.

3.2 Solution Requirement

Functional requirements

ID	Requirement
FR-1	Load the 100,000-record CSV and check field types and missing values.
FR-2	Show annual sales by manufacturer, by vehicle category and by model year.
FR-3	Show battery technology distribution and battery capacity against driving range.
FR-4	Show customer rating by category and price against customer rating.
FR-5	Show manufacturing distribution by country and CO savings by category.
FR-6	Combine all views in one published dashboard, grouped by scenario.

2

Non-functional requirements

Area	Requirement
Clarity	Clear titles, sorted axes, legends and tooltips on every view.
Accessibility	Viewable in a web browser through Tableau Public and an embedded project website.
Reproducibility	Dataset, workbook and website source stored in the GitHub repository.

3.3 Data Flow Diagram

```
Raw CSV (ev_market_test.csv, 100,000 rows x 37 columns)
      | import
      v
Tableau data source (field types and null values checked)
      | aggregate (SUM, AVG, COUNT) and group by dimension
      v
Nine worksheets
      | arrange by scenario
      v
Dashboard -->   Tableau Public  -->  Project website (GitHub Pages)
      |
      v
GitHub repository -->   SkillWallet review
```

3.4 Technology Stack

Tool	Use
Tableau Public	Data connection, worksheets and dashboard, published online.
CSV dataset / Microsoft Excel	Source data and basic inspection.
HTML and CSS	Project website that presents the overview, scenarios and embedded dashboard.
GitHub and GitHub Pages	Repository for all files and hosting for the project website.

4. Project Design

4.1 Problem Solution Fit

Scenario	Problem	How the dashboard fits
1. Market strategy director	Unclear which segments and manufacturers lead.	Sales by category, sales by model year and satisfaction by category show demand and quality side by side.
2. BI manager	No single view of competition and production geography.	Manufacturer sales, manufacturing distribution and price vs rating show competitive position.
3. Researcher	Unclear link between battery technology, range and sustainability.	Battery distribution, capacity vs range and CO_2 savings give the technology and impact evidence.

4.2 Proposed Solution

One Tableau dashboard with nine visualisations, published on Tableau Public and embedded in a project website.

No.	Visualisation	Scenario	Question answered
1	Annual EV Sales by Manufacturer	2 (and 1)	Who sells the most, and is the market concentrated?
2	Sales by Vehicle Category	1	Which segments drive demand?
3	EV Sales by Model Year	1	Is demand growing across model years?
4	Battery Technology Distribution	3 (and 2)	Which battery technologies dominate?
5	Battery Capacity vs Driving Range	3	Does more capacity give more range?
6	Customer Satisfaction by Category	1	Which segments satisfy customers most?
7	Manufacturing Distribution	2 (and 3)	Where are EVs produced?
8	Price vs Customer Rating	2	Do higher prices bring higher ratings?
9	CO Savings by Category $_2$	3	Which categories save the most $CO?_2$

4.3 Solution Architecture

Layer	Component	Role
Data	ev_market_test.csv	Single source of vehicle, sales, battery, rating and CO_2 data.
Preparation	Tableau data source	Field type checks and handling of missing values.
Analytics	Nine worksheets	Sum, average and distribution calculations by dimension.
Presentation	Tableau dashboard and project website	Views grouped by scenario, embedded in a web page.
Delivery	Tableau Public, GitHub, SkillWallet	Publishing, storage and review.

5. Project Planning & Scheduling

5.1 Project Planning

Phase	Activity	Output
1	Study the three scenarios and the dataset	List of questions each view must answer
2	Prepare the data source	Checked field types and null values
3	Build the nine worksheets	Nine named worksheets
4	Assemble and publish the dashboard	Tableau Public dashboard
5	Build the project website	index.html hosted on GitHub Pages
6	Documentation and video	Project report and explanation video
7	Submit	GitHub repository and SkillWallet submission

6. Functional and Performance Testing

6.1 Performance Testing

Data quality was checked before building the views. The dataset has 100,000 rows and 37 columns, which Tableau aggregates without difficulty.

Test	Result
Records / columns	100,000 / 37
Duplicate rows and duplicate vehicle IDs	0
Negative values in any numeric column	0
Missing values	9,511 cells in 5 columns (9,169 rows have at least one): battery cycle life 2,028; resale value 3,057; infotainment screen 1,506; annual sales 989; customer rating 1,931
Model years	2018 to 2026
Customer rating range	1.75 to 5.0
Price range	About USD 900 to USD 410,000

Tableau ignores null values in sums and averages, so missing sales and rating values do not distort averages, but totals exclude those records. Reference values computed directly from the CSV, used to validate dashboard figures:

Check	Expected value
Total annual sales (sum)	About 772.6 million units
Top category by sales	Two-Wheeler Scooter, about 85.0 million units (11.0%)
Battery share of vehicles (NMC / LFP / NCA / Sodium-Ion / Solid-State)	33.8% / 32.1% / 17.2% / 9.8% / 7.0%
Correlation of battery capacity with range	About 0.27
Average customer rating	About 4.08 out of 5
Correlation of price with customer rating	About 0.00

7. Results

7.1 Key Findings

Scenario 1: Market Strategy Director

- Two-Wheeler Scooters lead category sales with about 85.0 million units (11.0% of total), followed by Sedans (78.3 million, 10.1%), Compact SUVs (76.1 million, 9.9%) and Hatchbacks (76.1 million). Electric Trucks, Pickup Trucks and Electric Buses are the smallest, each below 1% of sales.
- Sales by model year are flat: the yearly total stays between about 82.6 million (2025) and 88.1 million (2021), so no clear growth segment appears across model years.
- Customer satisfaction barely differs by category: every category averages between 4.07 and 4.10 out of 5.

Scenario 2: Business Intelligence Manager

- The market is highly fragmented. SurgeWorks leads with about 4.88 million units, yet the top 5 manufacturers together hold only 3.1% of sales and the top 10 hold 6.0%. No manufacturer dominates among the 200 in the data.
- Manufacturing is spread evenly across the 10 countries, each with 9.8% to 10.2% of sales. China is highest (10.2%) and France lowest (9.8%).

- Price does not predict customer rating (correlation about 0.00). Prices range from about USD 900 to USD 410,000, but ratings stay near 4.1 across the range.

Scenario 3: Sustainable Mobility Researcher

- NMC (33.8% of vehicles) and LFP (32.1%) together account for about two-thirds of the market. NCA holds 17.2%, Sodium-Ion 9.8% and Solid-State 7.0%.
- Battery capacity and range are only weakly related across the whole dataset (correlation about 0.27) because vehicle types differ so much: an Electric Bus averages about 300 kWh for about 270 km, while a Luxury Sedan averages about 90 kWh for about 549 km.
- Commercial vehicles save the most CO per vehicle: Pickup Trucks, Electric Trucks and Electric Buses average about 9.0 tons per year, against about 3.0 for most cars and about 1.6 for two- and three-wheelers. The dataset total is about 368,900 tons per year.

8. Advantages & Disadvantages

Advantages

- One dashboard answers the questions of three different stakeholders.
- Large dataset (100,000 records) with a wide set of dimensions: sales, battery, range, rating, geography and CO.
- Findings are checked against reference values from the raw data.
- Published on Tableau Public and embedded in a project website, so anyone can open it in a browser.

Disadvantages

- Values are spread almost evenly across manufacturers, model years, countries and categories, and ratings barely vary. The dataset appears simulated, so results describe patterns in this dataset and should not be treated as verified real-world statistics.
- About 9.2% of rows have at least one missing value, including 989 sales and 1,931 rating values.
- Category sales depend on how many records each category has, so commercial vehicle sales are naturally smaller.
- CO savings are per-vehicle estimates stored in the dataset, not measured values.
- Relationships shown (such as capacity and range) are associations, not proof of cause.
- Tableau Public dashboards are publicly visible.

9. Conclusion

The dashboard shows an EV market that is fragmented, with no manufacturer holding more than a small share, and with demand led by Two-Wheeler Scooters and mainstream cars rather than commercial vehicles. NMC and LFP are the two main battery technologies, while Sodium-Ion and Solid-State already hold a combined 16.8% of vehicles. Customer ratings are almost identical across categories and do not rise with price, and larger batteries do not automatically mean longer range. Commercial vehicles sell the least but save the most CO per vehicle. For the three stakeholders, the nine views turn 100,000 records into clear and comparable answers.

10. Future Scope

- Add real sales data over time to measure true growth by segment.
- Add charging infrastructure data to study how charging networks affect adoption.

- Add forecasting of sales and battery technology adoption.
- Use a regularly refreshed data feed instead of a fixed CSV.
- Add filters for region, manufacturer and model year on the dashboard.
- Use CO_2 lifecycle data to compare EVs with petrol and diesel vehicles.

11. Appendix

Item	Details
Source code	Project website source: index.html (HTML and CSS) in the GitHub repository. Tableau packaged workbook (.twbx) also stored in the repository.
Dataset	ev_market_test.csv in the GitHub repository
GitHub	https://github.com/visheshyadav007/EV_Market
Project website	https://visheshyadav007.github.io/EV_Market/#dashboard
Tableau Public dashboard	Open the published dashboard